

Measuring Production and Income

ECON 101 Macroeconomics

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Problem 1: Calculating GDP

A country produces the following goods and services in a year:

- 100 units of cars at \$20,000 each
- 500 units of bread at \$5 each
- 200 units of smartphones at \$500 each
- Services worth \$1,000,000

Calculate the Gross Domestic Product (GDP) of the country.

Solution to Problem 1

GDP is the total market value of all final goods and services produced within a country in a given period. It is calculated as:

$$\text{GDP} = \sum (\text{Quantity} \times \text{Price})$$

$$\text{GDP} = (100 \times 20,000) + (500 \times 5) + (200 \times 500) + 1,000,000$$

$$\text{GDP} = 2,000,000 + 2,500 + 100,000 + 1,000,000 = 3,102,500$$

Answer: GDP = \$3,102,500

Problem 2: GDP vs. GNP

A country has the following data:

- $\text{GDP} = \$10 \text{ billion}$
- Income earned by domestic residents from abroad = \$2 billion
- Income earned by foreigners in the domestic economy = \$1 billion

Calculate the Gross National Product (GNP).

Solution to Problem 2

GNP is the total income earned by a country's residents, including income from abroad. It is calculated as:

$$\text{GNP} = \text{GDP} + \text{Income from abroad} - \text{Income to foreigners}$$

$$\text{GNP} = 10 + 2 - 1 = 11$$

Answer: GNP = **\$11 billion**

Problem 3: Net Domestic Product (NDP)

A country has the following data:

- $\text{GDP} = \$15 \text{ billion}$
- $\text{Depreciation} = \$2 \text{ billion}$

Calculate the Net Domestic Product (NDP).

Solution to Problem 3

NDP is GDP minus depreciation (the wear and tear on capital goods). It is calculated as:

$$\text{NDP} = \text{GDP} - \text{Depreciation}$$

$$\text{NDP} = 15 - 2 = 13$$

Answer: NDP = \$13 billion

Problem 4: National Income

A country has the following data:

- $\text{NDP} = \$12 \text{ billion}$
- $\text{Indirect taxes} = \1 billion
- $\text{Subsidies} = \$0.5 \text{ billion}$

Calculate the National Income.

Solution to Problem 4

National Income is NDP minus indirect taxes plus subsidies. It is calculated as:

$$\text{National Income} = \text{NDP} - \text{Indirect taxes} + \text{Subsidies}$$

$$\text{National Income} = 12 - 1 + 0.5 = 11.5$$

Answer: National Income = **\$11.5 billion**

Problem 5: Personal Income

A country has the following data:

- National Income = \$20 billion
- Corporate taxes = \$3 billion
- Undistributed corporate profits = \$2 billion
- Social security contributions = \$1 billion
- Transfer payments = \$4 billion

Calculate the Personal Income.

Solution to Problem 5

Personal Income is National Income minus corporate taxes, undistributed profits, and social security contributions, plus transfer payments. It is calculated as:

Personal Income = National Income—Corporate taxes—Undistributed profits—

$$\text{Personal Income} = 20 - 3 - 2 - 1 + 4 = 18$$

Answer: Personal Income = **\$18 billion**

Problem 6: Disposable Income

A country has the following data:

- Personal Income = \$25 billion
- Personal taxes = \$5 billion

Calculate the Disposable Income.

Solution to Problem 6

Disposable Income is Personal Income minus personal taxes. It is calculated as:

$$\text{Disposable Income} = \text{Personal Income} - \text{Personal taxes}$$

$$\text{Disposable Income} = 25 - 5 = 20$$

Answer: Disposable Income = **\$20 billion**

Problem 7: Real GDP

A country has the following data:

- Nominal GDP in Year 1 = \$1,000 billion
- Nominal GDP in Year 2 = \$1,200 billion
- Price index in Year 1 = 100
- Price index in Year 2 = 110

Calculate the Real GDP in Year 2 using Year 1 as the base year.

Solution to Problem 7

Real GDP adjusts Nominal GDP for inflation using the price index. It is calculated as:

$$\text{Real GDP} = \frac{\text{Nominal GDP}}{\text{Price index}} \times 100$$

$$\text{Real GDP} = \frac{1,200}{110} \times 100 = 1,090.91$$

Answer: Real GDP = **\$1,090.91 billion**

Problem 8: GDP Deflator

A country has the following data:

- Nominal GDP = \$1,500 billion
- Real GDP = \$1,200 billion

Calculate the GDP deflator.

Solution to Problem 8

The GDP deflator measures the level of prices in the economy. It is calculated as:

$$\text{GDP Deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100$$

$$\text{GDP Deflator} = \frac{1,500}{1,200} \times 100 = 125$$

Answer: GDP Deflator = **125**

Problem 9: Per Capita GDP

A country has the following data:

- GDP = \$500 billion
- Population = 50 million

Calculate the GDP per capita.

Solution to Problem 9

GDP per capita is GDP divided by the population. It is calculated as:

$$\text{GDP per capita} = \frac{\text{GDP}}{\text{Population}}$$

$$\text{GDP per capita} = \frac{500,000,000,000}{50,000,000} = 10,000$$

Answer: GDP per capita = **\$10,000**

Problem 10: Expenditure Approach to GDP

A country has the following data:

- Consumption = \$400 billion
- Investment = \$150 billion
- Government spending = \$200 billion
- Exports = \$100 billion
- Imports = \$50 billion

Calculate the GDP using the expenditure approach.

Solution to Problem 10

The expenditure approach calculates GDP as the sum of consumption, investment, government spending, and net exports (exports minus imports). It is calculated as:

$$GDP = C + I + G + (X - M)$$

$$GDP = 400 + 150 + 200 + (100 - 50) = 800$$

Answer: GDP = \$800 billion